

CareerForge – AI-Powered Career Development Platform with DevOps, DevSecOps & MLOps Pipeline

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Abstract

CareerForge – AI-Powered Career Development Platform with DevOps, DevSecOps & MLOps Pipeline is a web-based platform designed to help students improve career readiness through AI-powered resume analysis, skill-gap identification, personalized learning recommendations, and interview preparation. The system provides secure authentication, user profiles, progress tracking, and an interactive career dashboard. Developed using Agile and Scrum practices, the platform integrates React.js, Spring Boot, and PostgreSQL. DevOps practices use Docker, GitHub Actions, and AWS for CI/CD and cloud deployment, while DevSecOps incorporates secure authentication and vulnerability scanning. Python, Scikit-learn, and MLflow support the machine learning lifecycle, creating a secure, scalable, and intelligent career development solution.

Problem Statement

Students often struggle with:

- Resume improvement
- Identifying skill gaps
- Choosing suitable career paths
- Finding relevant learning resources
- Preparing for technical interviews

Existing Challenge

Career preparation is often fragmented across multiple platforms, making it difficult for students to receive personalized and continuous guidance.

Objectives

- Develop an AI-powered career development platform
- Analyze resumes and identify relevant skills
- Identify skill gaps for target career roles
- Provide personalized career and learning recommendations
- Support interview preparation
- Track student career progress
- Apply Agile, DevOps, DevSecOps and MLOps practices

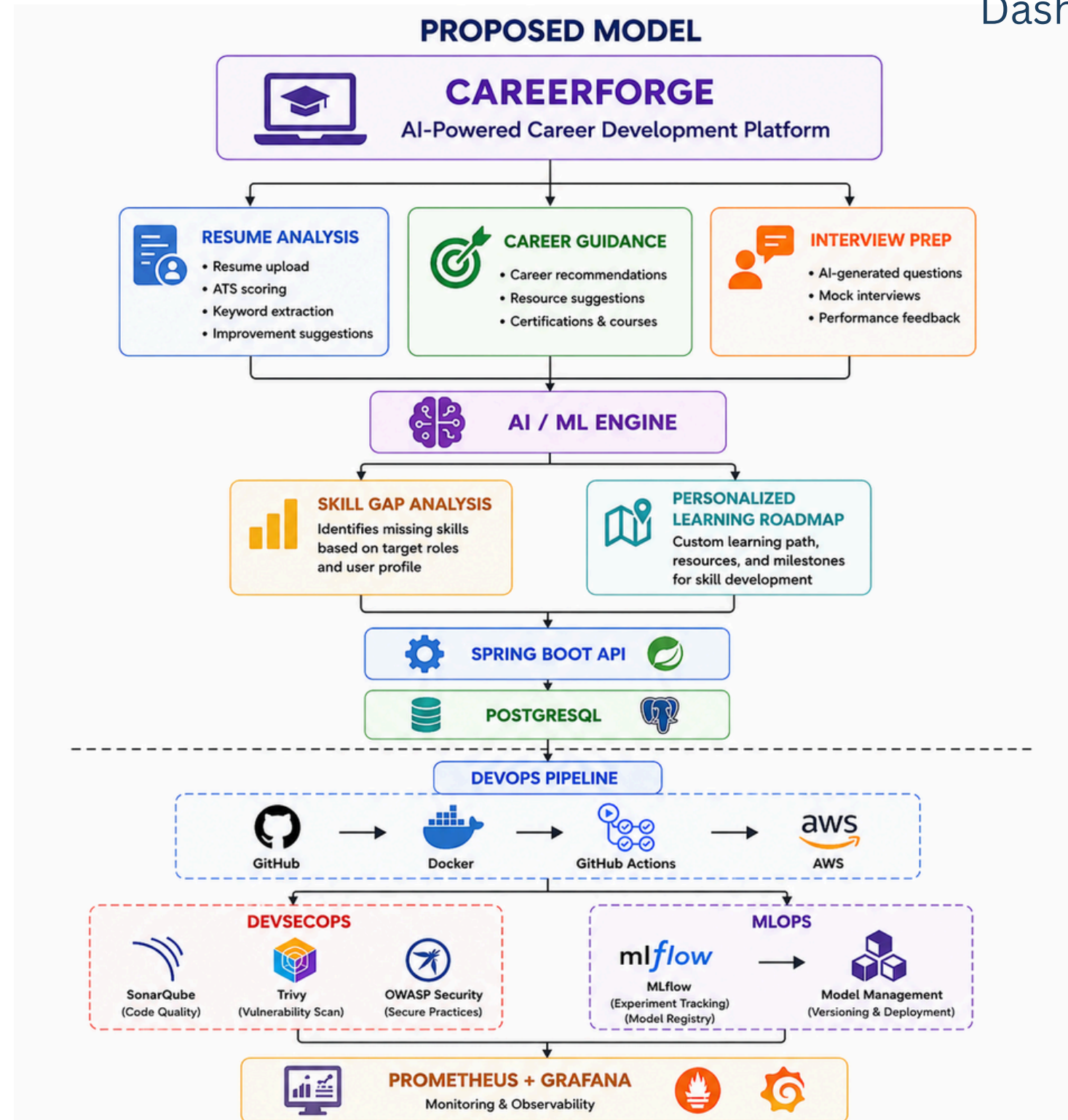
Existing Model

Existing System	Main Features	Limitation / Gap
<u>Jobscan</u>	ATS resume analysis, keyword matching, resume optimization	Mainly focused on resume/ATS optimization
<u>Teal</u>	AI resume builder, job matching, resume analysis	Strong career/job-search tools but limited DevOps/MLOps focus
<u>Career.io</u>	Resume optimization, job tracking, interview preparation	Broad career platform but doesn't focus on an integrated academic DevOps/DevSecOps/MLOps pipeline
<u>LinkedIn Career</u>	Job search, career guidance, AI-assisted career and resume features	Primarily a professional networking and recruitment platform
AI Career Recommendation Systems	Resume analysis, job-role prediction, skill-gap analysis, learning recommendations	Often research prototypes rather than complete production-oriented platforms

Proposed Model

CareerForge integrates career guidance, resume analysis, interview preparation, AI/ML, DevOps, DevSecOps, MLOps, and cloud deployment into a single platform.

User → Career Profile → AI Resume Analysis → Skill Gap Analysis → Personalized Career Roadmap → AI Interview Preparation → Progress Dashboard



Benefits

- Integrated career-development platform
- Personalized recommendations
- AI-assisted resume and skill analysis
- Structured learning guidance
- Continuous progress tracking
- Secure and scalable architecture

TECHNICAL ASPECTS

CareerForge integrates multiple technical components to deliver an **intelligent**, **secure** and **scalable** career development platform.



FRONTEND

- React.js with TypeScript
- Tailwind CSS for styling
- Responsive UI/UX
- Components for resume upload, dashboard, and career tracking



BACKEND

- Java 21, Spring Boot 3
- RESTful APIs
- Spring Security with JWT
- Role-based access control
- Business logic and service layer implementation



DATABASE

- PostgreSQL
- Stores users, resumes, skills, recommendations, interview data, and learning progress
- Designed for scalability and data integrity



AI / ML ENGINE

- Python, Scikit-learn, Pandas
- Resume parsing and skill extraction
- Skill-gap identification
- Career recommendations
- Interview question generation



DEVOPS

- Docker containerization
- GitHub for version control
- GitHub Actions for CI/CD
- Automated build, test and deployment pipeline
- AWS for cloud infrastructure



DEVSECOPS

- SonarQube for static code analysis
- Trivy for vulnerability scanning
- OWASP security best practices
- Secure coding & dependencies management



MLOPS

- MLflow for experiment tracking
- Model versioning and registry
- Model deployment
- Lifecycle management of ML models



MONITORING

- Prometheus for metrics collection
- Grafana for visualization and dashboards
- System and application performance monitoring
- Alerts and notifications



USERS
(Students / Admins)



WEB INTERFACE
(Frontend)



APPLICATION SERVER
(Backend APIs)



DATABASE
(PostgreSQL)



AI / ML ENGINE
(Python)



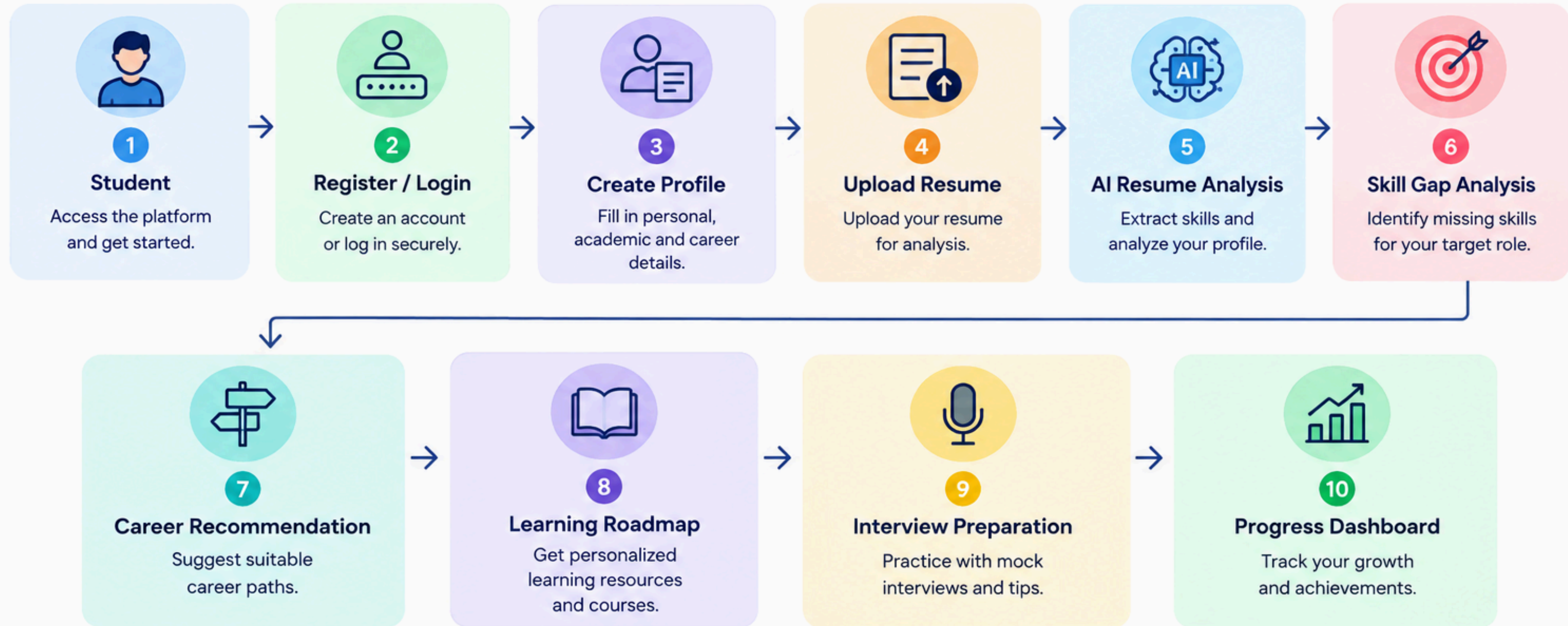
CLOUD & PIPELINE
(AWS, CI/CD, Security)



MONITORING
(Prometheus, Grafana)

CareerForge

System Workflow



Technology Stack



React.js
+ TypeScript
(Frontend)



Spring Boot
+ Java
(Backend)



PostgreSQL
(Database)



Python +
Scikit-learn + MLflow
(AI / ML)



Docker + GitHub Actions
+ AWS
(DevOps / Cloud)

Users

USERS

CareerForge has multiple user roles with distinct functionalities.



**STUDENT /
JOB SEEKER**

PRIMARY USER

-  Upload and analyze resumes
-  Identify skill gaps
-  Get personalized career recommendations
-  Practice AI-generated interview questions
-  Track career and learning progress



ADMINISTRATOR

SYSTEM MANAGER

-  Manage users and profiles
-  Manage interview questions and career resources
-  Monitor platform activity
-  View system analytics
-  Manage AI/ML-related resources



RECRUITER

FUTURE ENHANCEMENT

-  Post job/internship opportunities
-  Search candidates by skills
-  Review candidate profiles and resumes
-  Shortlist candidates



Students are the primary users of CareerForge.

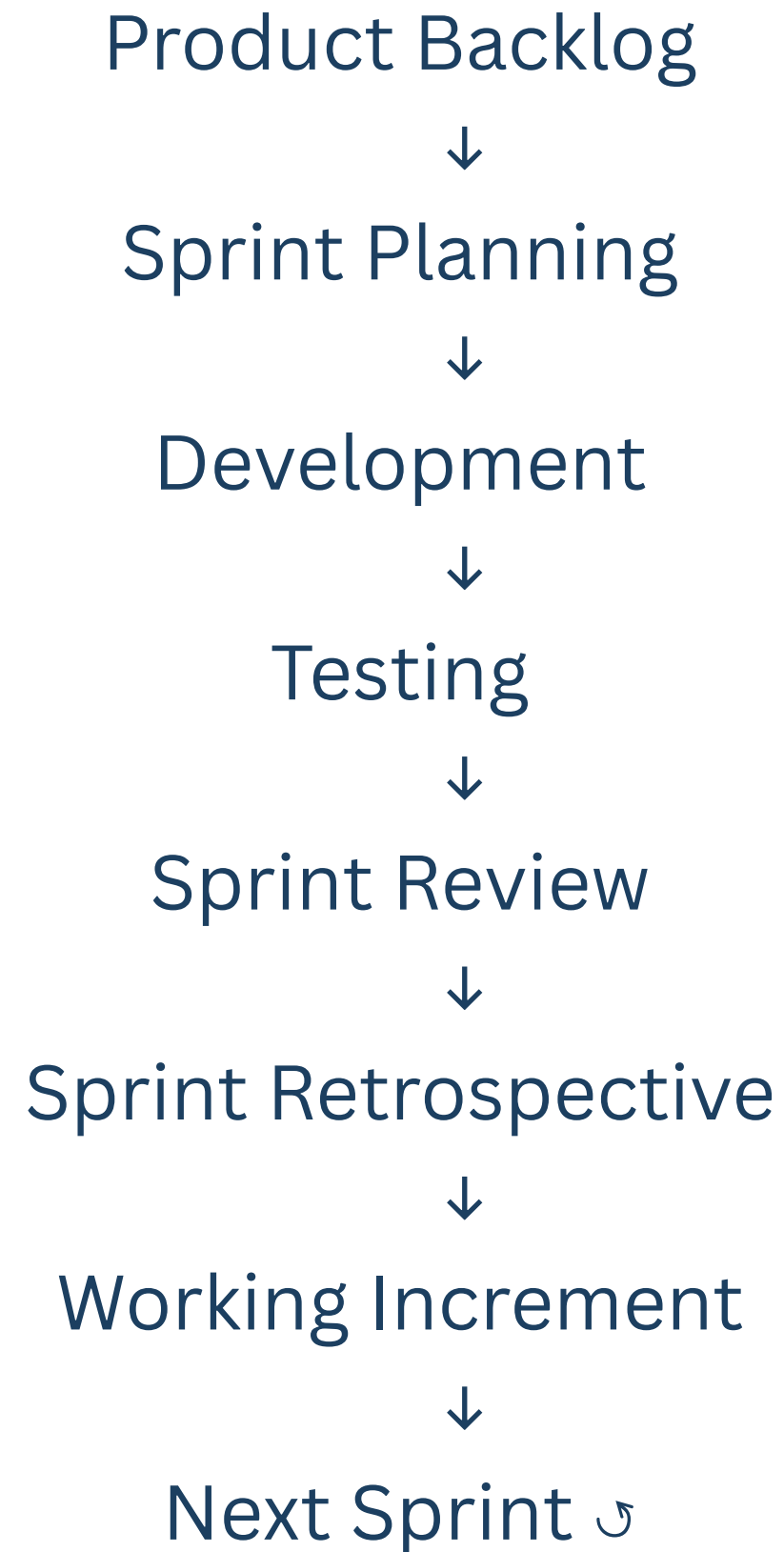
Administrators manage the platform, while **recruiters** will be added as a future enhancement.

Methodology



Process Model

Agile Scrum Model



Why Agile?

- Iterative development
- Continuous feedback
- Easy adaptation to changing requirements
- Frequent testing and improvement

- CareerForge provides an AI-powered platform for students to improve their career readiness.
- Integrates resume analysis, skill-gap identification, career recommendations, and interview preparation.
- Uses Agile Scrum for iterative and collaborative development.
- Implements DevOps and DevSecOps for automated, secure software delivery.
- Applies MLOps for machine-learning model management and deployment.
- Uses AWS and monitoring tools to support scalable and reliable deployment.

Conclusion

Thank You
